

MD5 Hash Collision in DocugamiReader Overwrites Structurally Distinct Chunks with Identical Text in run-llama/llama_index

✓ Valid

Reported on Apr 4th 2025

Description

The DocugamiReader class in llama_index retrieves structured XML documents from the Docugami API, parses them into semantic chunks, and converts them into Document objects. To assign consistent IDs to each chunk, the following logic is used:

```
_hashed_id = hashlib.md5(dg_chunk.text.encode()).hexdigest()
```

This hash is then used as a dictionary key:

```
framework_chunks[chunk_id] = framework_chunk
```

However, when two structurally distinct chunks contain the same text but appear in different parts of the document (i.e., different XPath), the same hash is produced. As a result, one chunk overwrites the other, leading to loss of contextually important information.

The following example shows a typical scenario. Identical text appears in two structurally distinct locations:

```
<Section>
  <Clause>This agreement may be terminated at any time.</Clause>
</Section>
<Appendix>
  <Note>This agreement may be terminated at any time.</Note>
</Appendix>
```

Although the content is the same, one is a binding clause while the other is a reference note. Since both receive the same MD5 hash, one chunk overwrites the other, and the legally binding clause is lost.

Proof of Concept

The following code replicates the structure used by DocugamiReader in llama_index. It shows how two document chunks with identical text but different XPath positions receive the same MD5

hash and how the latter silently overwrites the former.

```
from hashlib import md5
from dataclasses import dataclass
from typing import Optional, Dict, List

@dataclass
class FakeDocument:
    text: str
    metadata: Dict[str, str]
    excluded_llm_metadata_keys: Optional[List[str]] = None

XPATH_KEY = "xpath"
ID_KEY = "id"
DOCUMENT_NAME_KEY = "name"
STRUCTURE_KEY = "structure"
TAG_KEY = "tag"

@dataclass
class FakeChunk:
    text: str
    xpath: str
    structure: str
    tag: str
    parent: Optional['FakeChunk'] = None

document_name = "test_doc"
additional_doc_metadata = None
max_text_length = 4096
include_project_metadata_in_doc_metadata = True
parent_id_key = "doc_id"

def _build_framework_chunk(dg_chunk: FakeChunk) -> FakeDocument:
    _hashed_id = md5(dg_chunk.text.encode()).hexdigest()
    metadata = {
        XPATH_KEY: dg_chunk.xpath,
        ID_KEY: _hashed_id,
        DOCUMENT_NAME_KEY: document_name,
        STRUCTURE_KEY: dg_chunk.structure,
        TAG_KEY: dg_chunk.tag,
    }

    if additional_doc_metadata and include_project_metadata_in_doc_metadata:
        metadata.update(additional_doc_metadata)

    return FakeDocument(
        text=dg_chunk.text[:max_text_length],
        metadata=metadata,
        excluded_llm_metadata_keys=[XPATH_KEY, ID_KEY, STRUCTURE_KEY],
    )

parent_chunk = FakeChunk(
    text="Parent clause.",
    xpath="/Document/Section/Parent",
    structure="section",
    tag="section"
```

```

)

dg_chunks = [
    FakeChunk(
        text="This agreement may be terminated at any time.",
        xpath="/Document/Section/Clause",
        structure="clause",
        tag="p",
        parent=parent_chunk,
    ),
    FakeChunk(
        text="This agreement may be terminated at any time.",
        xpath="/Document/Appendix/Note",
        structure="note",
        tag="p",
        parent=parent_chunk,
    ),
]

framework_chunks: Dict[str, FakeDocument] = {}
for dg_chunk in dg_chunks:
    framework_chunk = _build_framework_chunk(dg_chunk)
    chunk_id = framework_chunk.metadata.get(ID_KEY)
    if chunk_id:
        framework_chunks[chunk_id] = framework_chunk
        if dg_chunk.parent:
            framework_parent_chunk = _build_framework_chunk(dg_chunk.parent)
            parent_id = framework_parent_chunk.metadata.get(ID_KEY)
            if parent_id and framework_parent_chunk.text:
                framework_chunk.metadata[parent_id_key] = parent_id
                framework_chunks[parent_id] = framework_parent_chunk

import pandas as pd

df = pd.DataFrame([
    {
        "id": doc.metadata[ID_KEY],
        "text": doc.text,
        "xpath": doc.metadata[XPATH_KEY],
        "structure": doc.metadata[STRUCTURE_KEY],
        "tag": doc.metadata[TAG_KEY],
        "is_parent": doc.text == "Parent clause.",
    }
    for doc in framework_chunks.values()
])

print(df.to_string(index=False))

```

Output

```

ID: 1feae597cd71576aa6ec7aaee67d0b10
Text: This agreement may be terminated at any time.
XPath: /Document/Appendix/Note

```

References

- [MD5](#)
- [Overwrite](#)

Impact

- Loss of semantically or legally important document content
- Breakage of parent-child chunk hierarchies
- Inaccurate or hallucinated responses in AI outputs

Occurrences

base.py L197-L208

⚙️ We are processing your report and will contact the [run-llama/llama_index](#) team within 24 hours. 10 months ago

🕒 A [GitHub Issue](#) asking the maintainers to create a `SECURITY.md` exists 10 months ago

Freedom of the Mind commented 9 months ago

Researcher

Hello, is the review of the above content in progress?

👁️ A [run-llama/llama_index](#) maintainer has acknowledged this report 8 months ago

📅 The scheduled publication date was automatically extended from 3rd Jul 2025 to 10th Jul 2025 due to the maintainers acknowledgement of the report 8 months ago

Clelia (Astra) Bertelli commented 8 months ago

Maintainer

Hey there, thanks for reporting this.

A [PR](#) is out to fix this issue

✅ **Logan** validated this vulnerability 7 months ago

💰 **freedom-of-the-mind** has been awarded the disclosure bounty ✅

💰 The fix bounty is now up for grabs

★ The researcher's credibility has increased: +7

 CVE-2025-6211 assigned to this report. 7 months ago

 Logan marked this as fixed in 0.3.1 with commit 29b2e0 7 months ago

\$ Clelia (Astra) Bertelli has been awarded the fix bounty 

\$ base.py#L197-L208 has been validated 

✉ We have notified the run-llama/llama_index maintainers about this report in their weekly follow-up 7 months ago

⚠ We have sent a warning to the run-llama/llama_index team to inform them that this report will be published in 48 hours 7 months ago

 This vulnerability has now been published 7 months ago

 CVE-2025-6211 has now been published 7 months ago

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CVE

CVE-2025-6211

Published

Vulnerability Type

CWE-440: Expected Behavior Violation

Severity

Medium (6.5)

Attack vector	Network
Attack complexity	Low
Privileges required	None
User interaction	None
Scope	Unchanged
Confidentiality	None
Integrity	Low
Availability	Low

[Open in visual CVSS calculator](#) 

Registry

Pypi

Affected Version

~ latest(v0.12.28)

Visibility

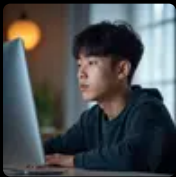
Public

Status
Fixed

Disclosure Bounty
\$125

Fix Bounty
\$31.25

Found by



Freedom of the Mind
@freedom-of-the-mind
MIDDLEWEIGHT



Fixed by



Clelia (Astra) Bertelli
@astrabert
UNPROVEN



Supported by [Palo Alto Networks](#) and Prisma AIRS, leading the way to greater AI security.

